Ex:- $\sqrt[3]{19683}$

Step1:- observe Qs unit digit i.e., '3'

Step2:- being 3 as unit digit, Qs cube root ends with '7'

Step3:- Analyse Qs cube root Qs Qn between -

Ie $20^3 = 8000$, $30^3 = 27,000$

So, Qs cube root Qs Qn between 20 & 30

Step4:- the number which ends with '7' and Qn b/w 20 & 30 Qs '27'.

$\therefore \sqrt[3]{19683} = 27$

(xi) knowing squares from 1 to 30:-

(xii) knowing cubes from 1 to 15:-

(xiii) General Multiplication techniques:-

i) Devide & Rule policy:-

Ex:- $73 \times 21$

Step1:- $73 \times (20+1)$ (or) $21 \times (70+3)$

Step2:- $73 \times 20 + 73 \times 1$ (or) $21 \times 70 + 21 \times 3$

Step3:- $1460 + 73$ (or) $1470 + 63$
[(70+3)20] [(20+1)70]

Step4:- 1533 (or) 1533

Ex:- $87 \times 23 = 87 \times 20 + 87 \times 3$
$= 1740 + 261$
$= 2001$

② Multiplied by 25/250/2500...

Ex:- $76 \times 25 = \frac{19}{76} \times \frac{100}{4} = 1900$

③ Multiplied by 50/500/5000...

Ex:- $84 \times 500 = \frac{42}{84} \times \frac{1000}{2} = 4200$

(xiv) Square of a number if we know its next (or) previous square value:-

Ex:- 1:- $39^2$ lets consider $x = 39$

for this we know $40^2 = 1600$ & its resultant is $< 40^2$.

So, that

$$40^2 - 40 - 39 = 39^2 = 1600 - 40 - 39$$
$$= 1560 - 39 = 1521$$
$$x^2 = (x+1)^2 - (x+1) - x$$

Ex2:- $81^2$ assume $x = 81$

for this we know $80^2 = 6400$ and $81^2$ is $> 80^2$. So,

$$81^2 = 80^2 + 80 + 81 = 6400 + 80 + 81$$
$$= 6561$$
$$\therefore x^2 = (x-1)^2 + (x-1) + x$$

(xv) Basic logic Behind squares of Natural No.s:-

$1^2 = 1$ ] +3
$2^2 = 4$ ] +5
$3^2 = 9$ ] +7
$4^2 = 16$ ] +9
$5^2 = 25$ ] +11
$6^2 = 36$

(i) Consecutive squares of natural No.s are differed with consecutive odd no.s.

(ii) $4^2 = 3^2 + 4^{th}$ odd number
$= 9 + (4a - 1) = 9 + (4 \times 2 - 1) = 16$

$\therefore n^2 = (n-1)^2 + n^{th}$ odd number.